

Design and Verification of a Two-Chip-Cascaded 4D mm-Wave Imaging Radar for Indoor SLAM

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Abstract — Since the concept of simultaneous localization and mapping (SLAM) was proposed and investigated in 1980s, it has been attracting continuous attention in robotics regime. The sensor types applied for perception in SLAM are usually stereo or RGB cameras, also lidars based on point clouds output with high density and accuracy. Driven by the autonomous driving in the automobile industry emphasizing both performance and cost, the 4D millimeter-wave imaging radar becomes a new alternative for the SLAM perception. This work focuses on a SLAM oriented solution merely using the implemented mm-wave 4D imaging radar for an indoor demonstration. The two-chip cascaded 4D radar offers totally 6 transmitting and 8 receiving physic channels in RF front-end. By applying sparse antenna array and multi-input multi-output (MIMO) technique, 48 virtual channels can be emulated, which achieve angle separation each of 1.5° on the azimuth and 3° on the elevation plane. An indoor 3D SLAM demonstration has been also done based on real-time point clouds output of the implemented 4D imaging radar.

Keywords — SLAM, 4D, mm-wave, imaging radar, sparse antenna array, MIMO.

I. INTRODUCTION

The concept of simultaneous localization and mapping (SLAM) was originally proposed during the IEEE Robotics and Automation Conference in 1986, San Francisco, California [1]. Since then, SLAM has become one of the most challenging but fascinating research topics in robotics regime. SLAM solution endues a robot the capability to plan its own path in an autonomous manner.

Camera -based SLAM have paved the way with great success as virtual method dominates in robotics perception systems [2]- [7]. Despite high cost, lidar-based SLAM [8]- [11] also becomes as one promising approach due to recently emerging lidar application in autonomous driving functions. As the conventional advanced driver assistant system (ADAS) millimeter-wave (mm-wave) radar has begun its evolution towards 4D point cloud imaging phase [12]- [14], mm-wave radar-based SLAM solutions have been correspondingly brought under spotlight [15]-[17] to enhance system performance under bad weather condition (rainy, snowy, foggy...etc.) in autonomous driving functions. This work presents the design, implementation, and verification of a two-chip cascaded 4D mm-wave imaging radar, which is suitable for an indoor SLAM application. By applying sparse antenna array and multi-input multi-output (MIMO) technique, 48

virtual channels can be emulated based on physically existing 6 transmitting and 8 receiving channels, which achieve angle separation each of 1.5° on azimuth and 3° on elevation plane. With maximum detection range of 51 m, the implemented 4D radar can demonstrate reasonable imaging effect for SLAM application by generating 10240 points/s in various indoor scenarios. The demonstration showed that the implemented 4D mm-wave imaging radar could be well applied for pure radar-based indoor 3D SLAM applications.

II. SYSTEM DESIGN

A. System Block Diagram

Figure 1 shows the system block diagram of the implemented 4D radar sensor. It consists mainly of two TEF8232 radar transceiver front-end chips and S32R294 microprocessor of power PC architecture. Since two radar transceiver chips each with 3 transmitting and 4 receiving channels are cascaded in form of “Master” and “Slave” by sharing the common local oscillator (LO) and clock (CLK) signal, the overall system owns then synchronized 6 transmitting and 8 receiving channels.

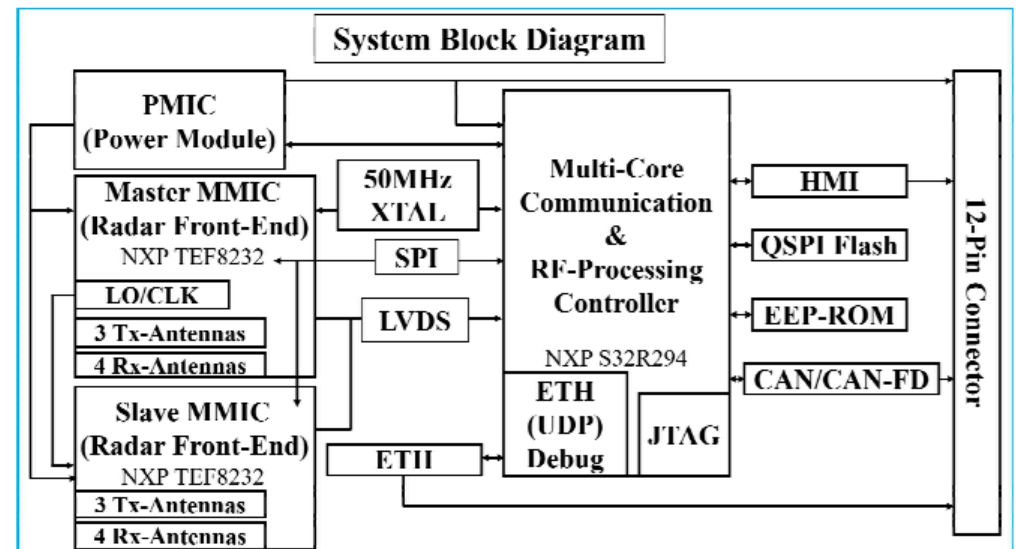


Fig. 1. System block diagram of the implemented two-chip-cascaded 4D mm-wave imaging radar sensor mainly based on two cascaded radar front-end chips (TEF8232) and one processing unit (S32R294).

B. Sparse Antenna Array

Figure 2 presents the antenna array design of the implemented 4D imaging radar sensor. With MIMO and sparse array technique, 6 transmitting antennas (Tx) and 8 receiving antennas (Rx) are so arranged on the printed circuits board

用於室內 SLAM 的雙片層疊式 4D 毫米波成像雷達的設計與驗證

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摘要 - 自 1980 年代提出並研究同步定位與繪圖 (SLAM) 概念以來，此概念在機器人領域一直備受關注。在 SLAM 中，用於感知的感測器類型通常是立體或 RGB 攝影機，也有基於高密度和高準確度點雲輸出的激光雷達。在強調效能與成本的汽車產業自動駕駛驅動下，4D 毫米波成像雷達成為 SLAM 感知的新選擇。本工作的重點在於以 SLAM 為導向的解決方案，僅使用已實作的毫米波 4D 成像雷達進行室內示範。雙晶片層疊式 4D 雷達在射頻前端提供了 6 個發射與 8 個接收的物理通道。透過稀疏天線陣列和多輸入多輸出 (MIMO) 技術，可模擬 48 個虛擬通道，每個通道可達到方位角 1.5° 和仰角 3° 的角度距離。此外，還利用 4D 成像雷達輸出的即時點雲，進行了室內 3D SLAM 演示。

關鍵字 - SLAM、4D、毫米波、成像雷達、稀疏天線陣列、MIMO。

以現有的 6 個發射與 8 個接收通道為基礎，模擬出 48 個虛擬通道，方位角與仰角各達 1.5°。所實作的 4D 雷達最大偵測距離為 51 公尺，可在各種室內情境下產生 10240 點/秒的速度，為 SLAM 應用展現合理的成像效果。本實驗結果顯示，所實作的 4D 毫米波成像雷達可以很好地應用於純雷達式的室內 3D SLAM 應用。

II.系統設計

A.系統方塊圖

圖一為 4D 雷達感測器的系統方塊圖。它主要由兩顆 TEF8232 雷達收發器前端晶片和 S32R294 微處理器組成的 Power PC 架構。由於兩顆雷達收發器晶片各擁有 3 個發射與 4 個接收通道，透過共用本機振盪器 (LO) 與時鐘 (CLK) 訊號，以「主」與「從」的方式串聯，整體系統便擁有 6 個同步發射與 8 個同步接收通道。

I.引言

同步定位與對應 (SLAM) 的概念最初是在 1986 年加州舊金山舉行的 IEEE 機器人與自動化會議上提出 [1]。從那時起，SLAM 已經成為機器人系統中最具挑戰性但又最吸引人的研究主題之一。SLAM 解決方案讓機器人有能力自主規劃路徑。

以攝影機為基礎的 SLAM 在機器人感知系統 [2]- [7] 中佔據主導地位，因此已成功鋪平了道路。儘管成本高昂，基於雷達的 SLAM [8]- [11]也因最近新興的雷達應用在自動駕駛功能上而成為一種有前途的方法。隨著傳統先進駕駛輔助系統 (ADAS) 毫米波雷達 (mm-wave) 開始向 4D 點雲成像階段演進 [12] - [14]，基於毫米波雷達的 SLAM 解決方案也相應地受到關注 [15] - [17]，以提高自主駕駛功能中惡劣天氣條件 (雨、雪、霧.....等) 下的系統性能。本研究提出一種適用於室內 SLAM 應用的雙晶片層疊式 4D mm 波成像雷達的設計、實作與驗證。透過應用稀疏天線陣列和多輸入多輸出 (MIMO) 技術

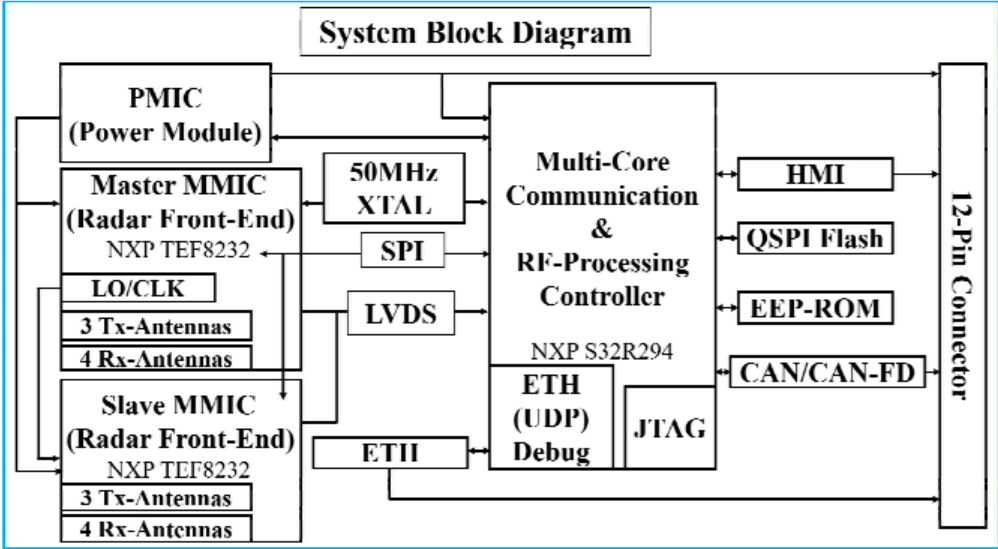


圖 1.主要基於兩個層疊式雷達前端晶片 (TEF8232) 和一個處理單元 (S32R294) 所實現的雙片層疊式 4D 毫米波成像雷達感測器的系統方塊圖。

B.稀疏天線陣列

圖 2 顯示 4D 成像雷達感測器的天線陣列設計。利用 MIMO 和稀疏陣列技術，6 個發射天線 (Tx) 和 8 個接收天線 (Rx) 如此排列在印刷電路板上。

(PCB) in Fig. 2 (left), that the equivalent virtual antenna array can achieve the maximum aperture each of 25.5λ on azimuth and 19.25λ on elevation plane with λ of the wavelength in 77/79 GHz frequency band.

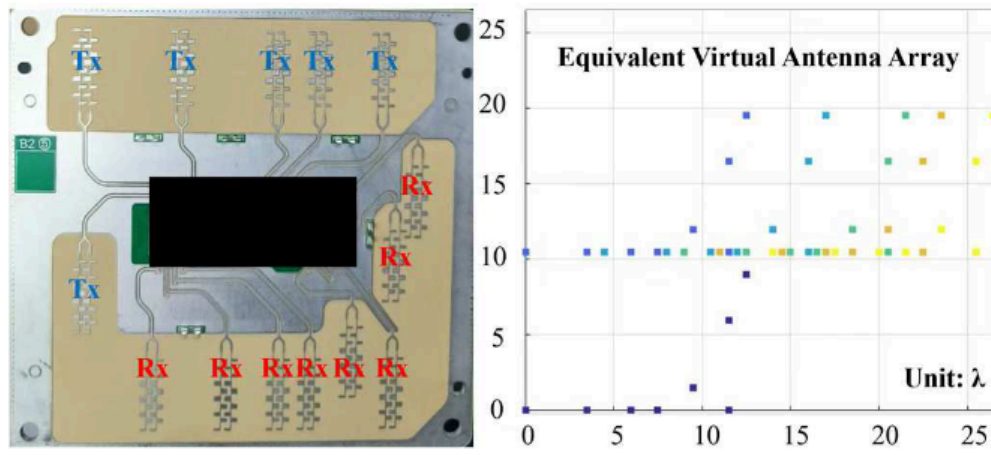


Fig. 2. Sparse array antenna distribution on implemented PCB with 6 transmitting and 8 receiving channels (left); equivalent virtual antenna array with 48 channels distributed on azimuth and elevation plane (right).

This implementation distributes all 48 virtual channels based on the following two principles: 1. Aperture on both azimuth and elevation plane should be larger than 15λ but less than 27λ (limited by the total radar size); 2. Angular separation of the implemented 4D imaging radar sensor on both azimuth and elevation plane should achieve 1.5° and 3° .

C. Mechanical Design Concept

The explosion view of the implemented 4D imaging radar sensor is shown in Fig. 3 (left) with the detailed dimension information in Fig. 3 (right). Millimeter-wave components such as cascaded radar chips, antenna array, power distribution networks, crystal are located on high frequency (HF) board. Processor chip, power management module, and all peripherals (flash, EEP-ROM, Ethernet, CAN, HMI...etc.) are populated on low frequency (LF) board.

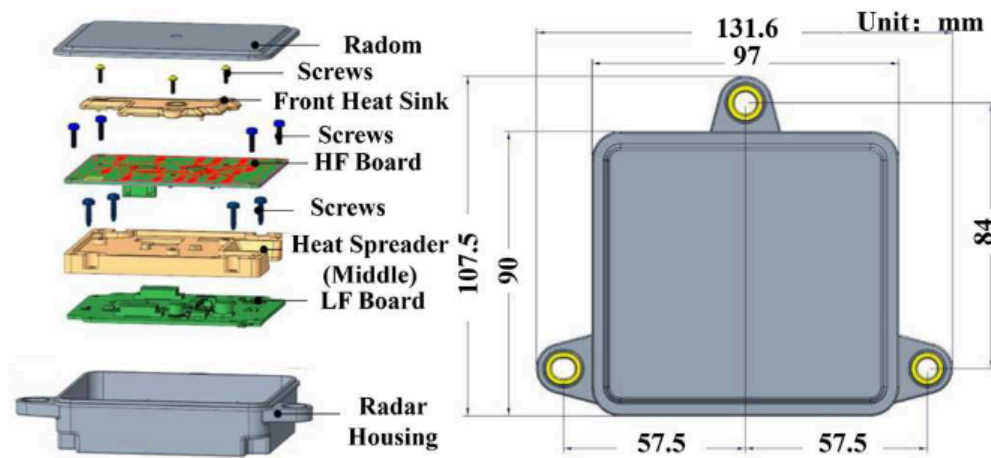


Fig. 3. Mechanical concept of the implemented 4D mm-wave imaging radar sensor (explosion view: left; dimensions: right).

Table 1. Angular resolution measurement on the azimuth and the elevation planes for the implemented 4D mm-wave imaging radar.

Set Value			4D Radar Under Test (RUT)	
Angle/Plane	Target RCS	Range	Distance of two corner reflectors	Angular separation
0° /Az.	10 dBsm	10 m	26.88 cm	1.44°
0° /El.	10 dBsm	10 m	50.28 cm	2.88°

Thermal management considered design is implemented via the front heat sink and the heat spreader between HF and LF boards. Meantime the assembly process of the HF/LF boards and the heat spreader in the middle enables a sufficient electromagnetic compatibility (EMC). The entire sensor has a compact size of $131.6 \text{ mm} \times 107.5 \text{ mm} \times 25.45 \text{ mm}$, which is reasonably convenient for integration into various robotic platforms.

III. VERIFICATION RESULTS

A. Angular Resolution Test

To verify the angular separation capability of the implemented 4D radar sensor, two corner reflectors with radar cross section (RCS) value of 10 dBsm were placed each on the both sides along the boresight of the radar under test (RUT) in an anechoic chamber. The distance between RUT and two corner reflectors was set to 10 m with a calibrated optical bench. The minimum distance between two corner reflectors was noted, at which the RUT could still distinguish two corner reflectors as two separated groups of target points. The process was repeated for the test cases each on the azimuth and the elevation planes. Table 1 presents the tested angular separation results for the mentioned test cases. The designed angular separation of 1.5° on the azimuth plane and 3° on the elevation plane have been well realized.

B. Field-of-View (FoV)

To verify the FoV of the implemented 4D radar sensor, static targets with various RCS values of -5dBsm, 0 dBsm, 10 dBsm, and 20 dBsm were set for extracting maximum detection range at various incident angles during the measurement. The maximum detection range values were noted only when a stable detection behavior was confirmed (at least 90% detection statistics verified by the data flow). The verified FoV diagrams each of azimuth and elevation plane are presented in Fig. 4 with accordingly RCS values of -5dBsm, 0 dBsm, 10 dBsm, and 20 dBsm, respectively.

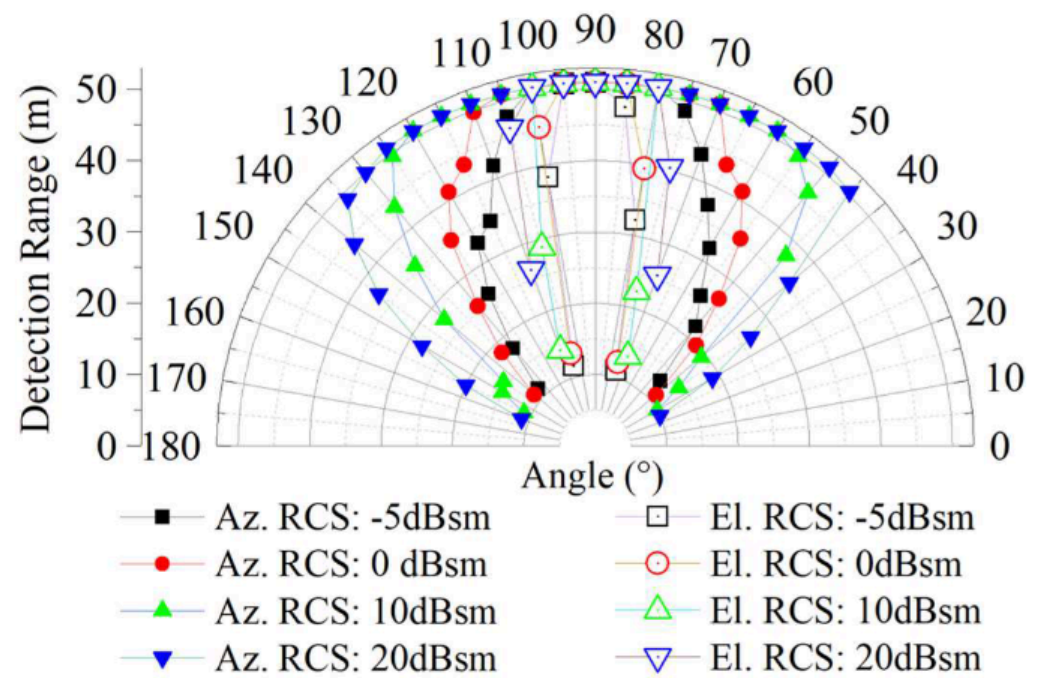


Fig. 4. Tested FoV of the implemented 4D mm-wave imaging radar sensor each on the azimuth (blue square point-line) and on the elevation (red triangle point-line) plane using a target with RCS values of -5 dBsm (black), 0 dBsm (red), 10 dBsm (green), and 20 dBsm (blue).

(PCB)，等效虛擬天線陣列在 77/79 GHz 頻段可達到方位角 25.5λ 和仰角 19.25λ 的最大孔徑，波長為 λ 。

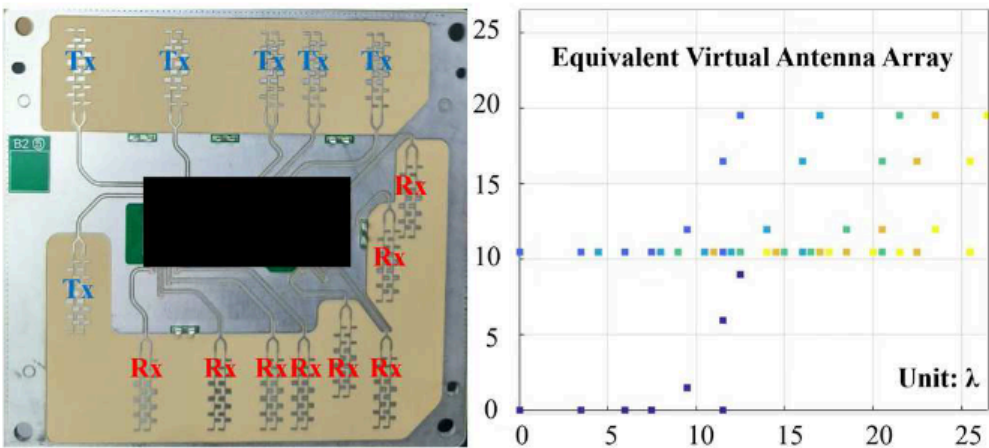


圖 2.6 個傳送通道和 8 個接收通道的稀疏陣列天線在實作 PCB 上的分佈 (左圖)；48 個通道分佈在方位角和仰角平面上的等效虛擬天線陣列 (右圖)。

此實作根據以下兩項原則分佈所有 48 個虛擬通道：1.方位和仰角平面上的孔徑應大於 15λ 但小於 27λ (受雷達總尺寸限制)；2.實施的 4D 成像雷達感測器在方位和仰角平面上的角度分離應達到 1.5° 和 3° 。

C.機械設計概念

圖 3 (左) 為 4D 成像雷達感測器的爆炸視圖，圖 3 (右) 為詳細尺寸資訊。毫米波元件如層疊式雷達晶片、天線陣列、配電網路、晶體等位於高頻 (HF) 板上。處理器晶片、電源管理模組和所有周邊設備 (快閃記憶體、EEP-ROM、乙太網路、CAN、人機介面...等) 則安裝在低頻 (LF) 板上。

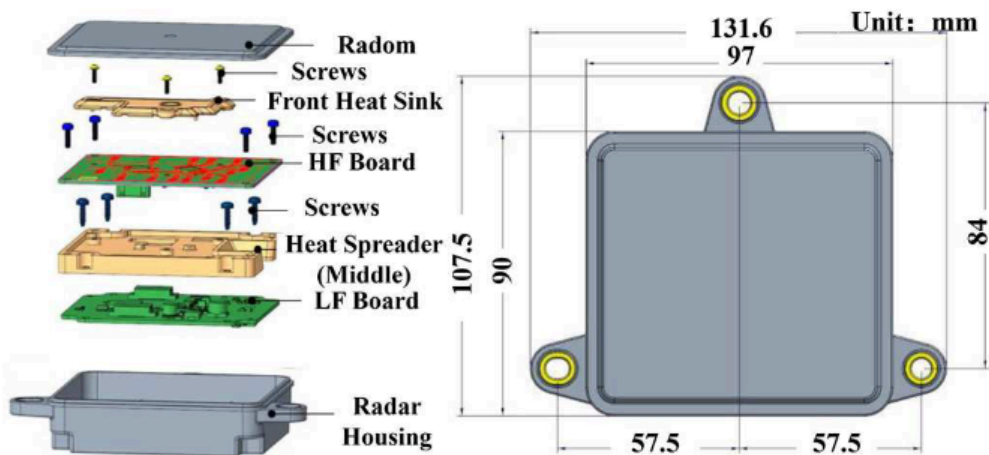


圖 3.實現的 4D 毫米波成像雷達感測器的機械概念（爆炸視圖：左；尺寸：右）。

表 1.4D 毫米波成像雷達在方位角和仰角平面上的角度解析度測量結果。

設定值			測試中的 4D 雷達 (RUT)	
角度/平面目標	RCS	範圍	距離 兩角 反光罩	角度 分離
0°/Az.	10 dBsm	10 m	26.88 公分	1.44°
0°/El.	10 dBsm	10 m	50.28 公分	2.88°

透過前端散熱器和高頻與低頻板之間的散熱器，實現了考慮周全的熱能管理設計。同時，HF/LF 板的組裝過程以及中間的散熱器使其具有足夠的電磁相容性 (EMC)。整個感測器的尺寸為 $131.6\text{ mm}\times107.5\text{ mm}\times25.45\text{ mm}$ ，相當小巧，方便整合至各種機器人平台。

III.驗證結果

A.角度解析度測試

為了驗證 4D 雷達感測器的角度分離能力，在電波暗室中，沿被測雷達 (RUT) 的孔徑兩側各放置兩個雷達截面 (RCS) 值為 10 dBsm 的角反射器。RUT 與兩個角反射器之間的距離使用校準光學工作台設定為 10 公尺。記下兩個角反射器之間的最小距離，在此距離下 RUT 仍可將兩個角反射器區分為兩組不同的目標點。方位角和仰角平面上的測試案例均重複此程序。表 1 列出了上述測試案例的測試角度分離結果。方位角平面上 1.5° 和仰角平面上 3° 的設計角度分離已經很好地實現。

B.視場 (FoV)

為了驗證 4D 雷達感測器的 FoV，在量測過程中設定了不同 RCS 值 (-5dBsm、0 dBsm、10 dBsm 和 20 dBsm) 的靜態目標，以擷取不同入射角的最大偵測範圍。只有在確認穩定的偵測行為 (經資料流驗證至少有 90% 的偵測統計值) 時，才會記下最大偵測範圍值。圖 4 所示為方位角與仰角平面經驗證的 FoV 圖，其相對應的 RCS 值分別為 -5dBsm、0 dBsm、10 dBsm 及 20 dBsm。

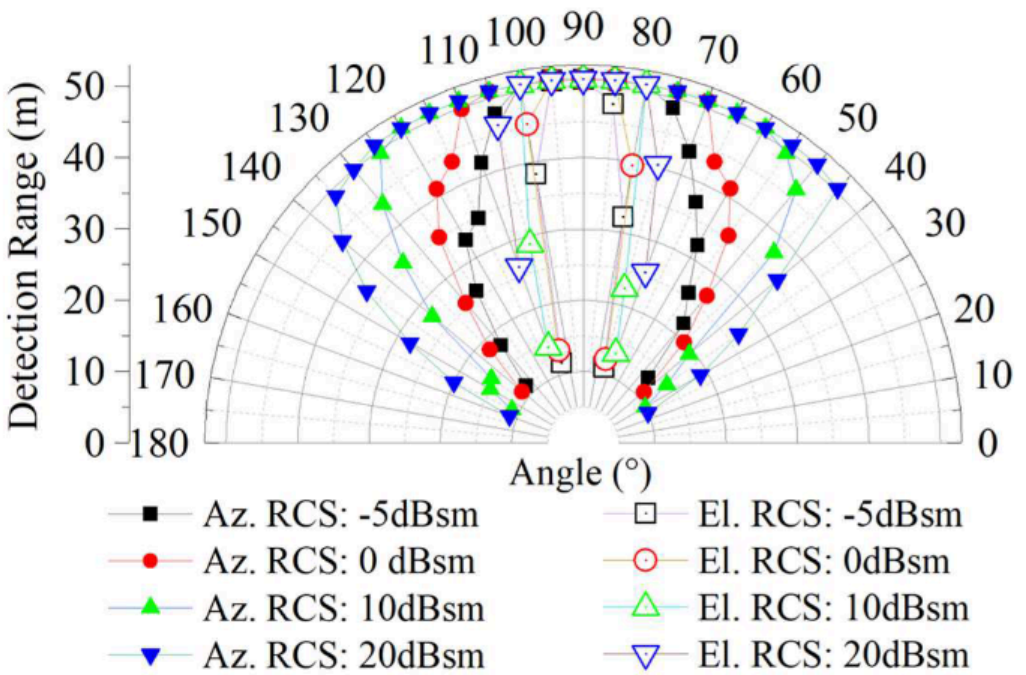


圖 4.使用 RCS 值為 -5 dBsm (黑色)、0 dBsm (紅色)、10 dBsm (綠色) 及 20 dBsm (藍色) 的目標，分別在方位角 (藍色方形點線) 及仰角 (紅色三角形點線) 平面上測試 4D 毫米波成像雷達感測器的 FoV。

C. Indoor SLAM Demonstration

The feasibility of indoor radar-based SLAM was demonstrated in a proof-of-concept scenario using an indoor three-wheel hospitality robot as show in Fig. 5. The robot was remote controlled by a test engineer with a typical speed of 1-2 m/s. The indoor test location was selected to resemble the typical features of industrial workshops (metal shelves, industrial machines, pillars) and tradeshow halls (chairs, podiums, tall tables) to address potential future use cases.



Fig. 5. Demonstration scenario and setup of the indoor radar-based SLAM using a hospitality robot platform.

The robot was equipped with a stereo RGB camera (30 Hz), a solid-state scanning lidar (10 Hz), and three in this work implemented 4D radar sensor prototypes (10 Hz) mounted in a way that one points along the front of the ego vehicle, while the other two to the rear-left and -right around $\pm 120^\circ$ respectively. This surround radar view ensures that at least one of the sensors have visibility over (a part of) the scene even when the robot is close to walls or other large objects on one side. All sensors were mounted on a metallic pole structure (Fig. 5) to give them better visibility over the chassis of the robot. All sensors were calibrated, synchronized based on network time protocol (NTP), and integrated into the robotic operating system (ROS) for easier sensor management, visualization, and data recording.

Since global navigation satellite system (GNSS) signal is not available indoors as reference, lidar-based SLAM was chosen to create the reference trajectories for the proposed radar-based SLAM solution. To do this, the widely used Kiss-ICP SLAM method [18] was applied on the output from the lidar sensor mounted on the test robot. While the results are plausible, it is worth mentioning that the trajectories obtained with this method should only be considered as references instead of ground truth. A good illustration of this can be seen in Fig. 6, where the lidar map is imperfect in the top left corner. Possible reasons for this could be the limited FoV of the used lidar sensor leading to improper point cloud registration when the sensor is too close to the walls or corners.

Indoor use of radar sensors is often limited by strong multipath effects caused by close-by, highly reflective vertical structures and the fact that such sensors are usually configured (maximum range, range resolution) for automotive driving scenarios. Thus, before applying algorithms to estimate the ego-motion of the sensor, it is critical to "clean" the radar point cloud as much as possible. To do this, filters were applied based

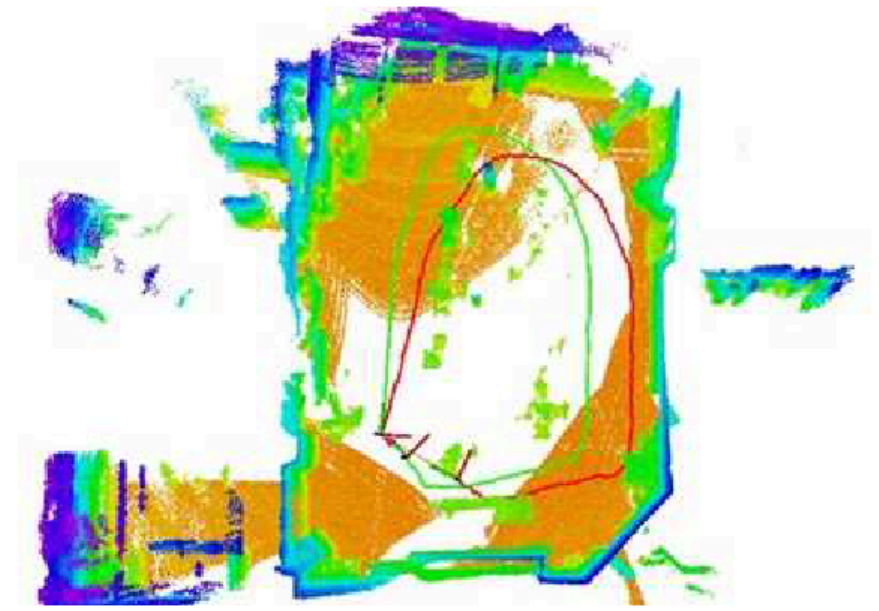


Fig. 6. Top view lidar map of the test scenario each with lidar- and radar-based trajectories marked with green (lidar-based) and red (radar-based) loops.

on distance (maximum 10 meters), reflectivity (minimum 35 signal-noise-ratio), and only points resulting from the peaks in the radar-cube are considered. Furthermore, all points were projected to a ground plane (i.e. assuming $Z = 0$) to simplify the regression of ego-motion. These parameters can be adjusted based on the target environment and the end use-case.

The scope of this work was to provide a proof-of-concept demonstration of the possibility of using 4D imaging radar sensors indoors for SLAM purpose. Thus, instead of developing novel methods, existing solutions were tried out on the pre-filtered radar point cloud. Given that the scenario only involved motion of the robot in planar environments without change in elevation, we assume zero translation in the vertical (Z axis) direction and zero roll and pitch for the ego-motion estimation. First, as a baseline, a basic point cloud matching using the Kabsch-Umeyama [19] algorithm between two consecutive scans is implemented. As seen in Fig. 7, the resulting trajectory (yellow) is far from the reference lidar (green) trajectory but indicates strong potential for point cloud matching approaches. Subsequently, the KISS-ICP [18] algorithm was modified to include the assumptions about ego-motion, i.e., only translation along longitudinal and lateral direction and yaw rotation is considered.

In Fig. 7, the red trajectory represents the output of this approach with much more accurate results than the baseline. Both methods are implemented on central processing units (CPUs) in real-time. As seen in Fig. 6 and Fig. 7, radar-based



Fig. 7. Qualitative comparison of the LiDAR-based reference trajectory (green), and the radar-based trajectories using basic point cloud matching with the Kabsch-Umeyama algorithm (yellow) and Kiss-ICP (red).

C.室內 SLAM 示範

如圖 5 所示，使用室內三輪接待機器人進行概念驗證，證明了室內雷達 SLAM 的可行性。該機器人由測試工程師遙控，典型速度為 1-2 m/s。室內測試地點的選擇與工業車間（金屬架、工業機器、支柱）和展覽大廳（椅子、講台、高腳桌）的典型特徵相似，以滿足未來潛在的使用個案。



圖 5.使用款待機器人平台的室內雷達式 SLAM 示範場景與設定。

機器人配備了立體 RGB 攝影機 (30 Hz)、固態掃描雷達 (10 Hz)，以及三個在本工作中實現的 4D 雷達感應器原型 (10 Hz)，安裝方式為其中一個沿著自我車輛的前方，而另外兩個則分別指向± 120°左右的後左方和右方。這種環繞雷達檢視方式可確保即使機器人靠近牆壁或其他大型物體的一側，至少有一個感測器能看到（部分）場景。所有感測器都安裝在金屬圓柱結構上（圖 5），讓它們在機器人的底盤上有更好的可視性。所有感測器都經過校準，並根據網路時間協定 (NTP) 進行同步，並整合到機器人作業系統 (ROS) 中，以方便感測器管理、視覺化和資料記錄。

由於室內沒有全球導航衛星系統 (GNSS) 訊號可作為參考，因此選擇以雷射雷達為基礎的 SLAM 來為所提出的雷達 SLAM 解決方案建立參考軌跡。為了做到這一點，我們將廣泛使用的 KissICP SLAM 方法 [18] 應用於安裝在測試機器人上的雷達感測器的輸出。雖然結果是可信的，但值得一提的是，使用此方法獲得的軌跡應僅被視為參考值，而非地面真實值。圖 6 就是一個很好的說明，圖中左上角的雷達圖並不完美。可能的原因是所使用的雷達感應器 FoV 有限，導致感應器太靠近牆壁或角落時，點雲登錄不當。

雷達感測器的室內使用通常會受到近距離、高反射垂直結構所造成的強多徑效應限制，而且此類感測器通常是針對汽車駕駛情境而配置（最大範圍、範圍解析度）。因此，在應用演算法來估計感測器的自動性之前，儘可能「淨化」雷達點雲是非常重要的。要做到這一點

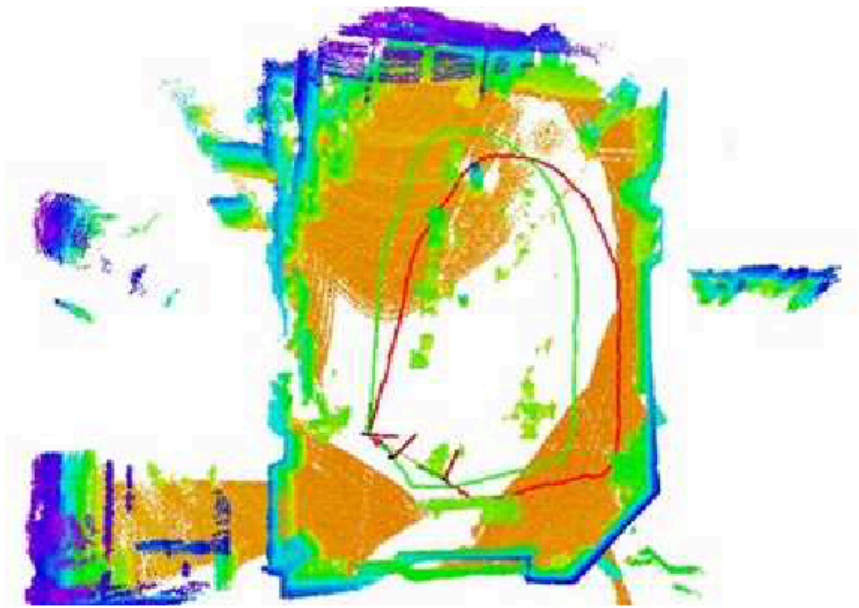


圖 6.測試場景的頂視圖雷達地圖，其中以綠色 (雷達) 和紅色 (雷達) 環標示出雷達和雷達軌跡。

根據距離 (最遠 10 公尺)、反射率 (信噪比最小 35)，並僅考慮雷達立方體中的峰值所產生的點來進行篩選。此外，所有點都投影到地平面（即假設 $Z = 0$ ），以簡化自我運動的回歸。這些參數可根據目標環境和最終使用情況進行調整。

本工作的範圍是提供概念驗證，證明在室內使用 4D 成像雷達感應器進行 SLAM 的可能性。因此，我們並沒有開發新的方法，而是在預濾雷達點雲上嘗試現有的解決方案。由於機器人僅在平面環境中運動，沒有高低變化，因此我們假設垂直（Z 軸）方向上的平移為零，而自我運動估計的滾動和俯仰均為零。首先，作為基線，在兩個連續掃描之間使用 Kabsch-Umeyama [19] 演算法執行基本的點雲比對。如圖 7 所示，結果軌跡 (黃色) 與參考雷達 (綠色) 軌跡相差甚遠，但顯示出點雲比對方法的強大潛力。之後，KISS-ICP [18] 演算法經過修改，加入了自我運動的假設，即只考慮沿縱向和橫向的平移以及偏航旋轉。

在圖 7 中，紅色軌跡代表此方法的輸出，其結果比基線精確得多。這兩種方法都在中央處理器上執行

(CPU) 實時運作。如圖 6 和圖 7 所示，以雷達為基礎的



圖 7.以 LiDAR 為基礎的參考軌跡 (綠色) 與使用 Kabsch-Umeyama 演算法 (黃色) 和 Kiss-ICP (紅色) 進行基本點雲比對的雷達軌跡的定性比較。

SLAM performs comparably to LiDAR-based SLAM for most parts of the trajectory. However, the performance drops in certain scenarios and motions. When the ego-vehicle is close to large artefacts and walls, the radar facing them cannot provide reliable point cloud output due to strong multipath and limited FoV. While the radars facing the other directions can help, the overall 360° point cloud becomes less salient for registration. This phenomenon is also observed while facing cornered walls where two out of the three sensors become momentarily quite noisy and thus the motion predicted during 90° turnings can be less accurate..

IV. CONCLUSION

A compact two-chip cascaded 4D imaging mm-wave radar sensor aiming for indoor 3D SLAM application is designed, implemented, and verified. By applying sparse antenna array and MIMO technique, 48 virtual channels are emulated, which can achieve angle separation each of 1.5° on azimuth and 3° on elevation plane. With maximum detection range of 51 m, the implemented 4D radar could well demonstrate a real-time mapping and localization effect in indoor scenarios. It is clear from the results that while adapting lidar-based solutions with minimal modifications gives promising results, there is room for improvement by addressing unique challenges and exploiting unique features offered by radars. First, in this work, the Doppler output of the radars was not utilized which could be a strong indicator for longitudinal translations and a signal to separate moving and static obstacles – a step critical in environments with humans. Second, it is well-known that the uncertainty in radars' angle estimation and radial velocity estimation depends on the angle-of-arrival and should be accounted for in the SLAM framework. Another consequence of these properties of radars is the poor rotation estimation, which could be addressed by introducing an inertial measurement unit (IMU). Finally, while the current setup with three radars provides a 360° view around the robot, it does not provide decent overlap between their field of views which could increase the robustness of selecting and matching stable points. In addition, the framework may also be adapted to ignore sensor(s) that are partially or fully blocked due to nearby walls, walking pedestrians, etc. leading to highly noisy point clouds. This allows for only stable and reliable radar detections to contribute to the localization and mapping efforts.

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在軌跡的大部分區域中，SLAM 的表現可媲美基於 LiDAR 的 SLAM。然而，在某些情況和動態下，性能會下降。當自我車輛接近大型假象和牆壁時，由於強大的多徑和有限的 FoV，面向這些假象和牆壁的雷達無法提供可靠的點雲輸出。雖然朝向其他方向的雷達可以提供協助，但整體 360° 點雲的註冊顯著度變得較低。當面對牆壁時也會出現這種現象，三個感應器中有兩個會暫時變得相當嘈雜，因此在 90° 轉彎時預測的運動會變得不太精確。

IV.設計、實作與驗證了一種針對室內 3D SLAM 應用的小型雙晶片層疊式 4D 成像毫米波雷達感測器。透過應用稀疏天線陣列與 MIMO 技術，模擬出 48 個虛擬通道，可達到方位角 1.5° 與仰角 3° 的角度距離。4D 雷達的最大偵測距離為 51 公尺，可在室內實現即時繪圖與定位的效果。從結果可以清楚看出，雖然以最小的修改來適配基於雷達的解決方案會帶來很好的結果，但仍有改善的空間，可以解決獨特的挑戰，並利用雷達所提供的獨特功能。首先，在這項工作中，雷達的多普勒輸出並未被利用，而多普勒輸出可作為縱向平移的強大指標，以及區分移動與靜止障礙物的信號 - 在有人影的環境中，這是非常重要的。其次，眾所周知，雷達的角度估計和徑向速度估計的不確定性取決於到達角，因此應在 SLAM 框架中加以考慮。雷達的這些特性所造成的另一個後果是較差的旋轉估計，這可以透過導入慣性測量單元 (IMU) 來解決。最後，雖然目前有三個雷達的設定可提供機器人周圍 360° 的視野，但卻無法在它們的視野間提供適當的重疊，而重疊可增加選擇和匹配穩定點的穩健性。此外，此架構也可以忽略因附近的牆壁、行走的行人等因素而部分或完全阻擋的感測器，導致高雜訊的點雲。如此一來，只有穩定可靠的雷達探測才有助於定位與繪圖工作。

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